One noisy cellular automaton with an uncomputable noise diagram

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Abstract

Can varying only the noise rate of one cellular automaton encode the halting problem? We construct one computably specified cellular automaton on $\mathbb{Z}^6$, with a finite alphabet and a finite neighbourhood, whose ergodicity under uniform replacement noise does so. There are pairwise disjoint open intervals I_n with rational endpoints computable from n such that, at every rate in I_n, the perturbed automaton is ergodic exactly when a fixed universal machine halts on input n. In the nonhalting case it has at least two translation-invariant stationary laws; in the halting case it has one stationary law attracting every initial law.

Finite rulers in a hierarchical tiling turn the noise rate into a cutoff in a sequence of local flags. Computation boards read that cutoff and simulate the selected input. A threshold checker preserves a global bit when the computation does not halt. When it halts, interval-valued sensors erase every surviving chart, and a common-noise coupling removes dependence on every initial state. The guaranteed intervals accumulate at zero and lie at astronomically small rates. The space–time Gibbs correspondence gives a finite-energy interaction on $\mathbb{Z}^7$ with undecidable translation-invariant Gibbs uniqueness. Thus one fixed local rule has an uncomputable noise diagram.

AI systems assisted with this work; see "How this work was done."

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1 Introduction

Can the long-run behaviour of one cellular automaton, subjected to the same noise mechanism at different rates, encode an undecidable problem? Independent replacement gives every symbol positive probability at every update. It nevertheless permits persistent information. We show that changing only the replacement rate can select which of two behaviours occurs: forgetting every initial state, or retaining a bit in two stationary laws.

Here is the main result in plain words. Fix a universal deterministic machine U whose halting set on positive unary inputs is undecidable.

Theorem A. There is one computably specified cellular automaton on $\mathbb{Z}^6$, with a finite alphabet and a finite neighbourhood, and computable disjoint rational open intervals I_n, such that at every rate in I_n uniform whole-symbol replacement makes the automaton ergodic exactly when $U(n)$ halts. Nonhalting gives at least two translation-invariant stationary laws. Halting gives a unique stationary law attracting every initial law.

In particular, the noise diagram, the set of rates at which F_ε is ergodic, is undecidable at rational rates for this rule (Corollary 9.4). The local rule receives no numerical noise parameter. Rulers of different lengths see different numbers of random replacements. Their responses select an integer, and finite computation boards read it. All boards use the same machine and the same alphabet. The computation is in the spatial constraints of a chart; a halting computation is excluded by forbidding its halting state. Figure 1 shows the two parts of this selection.

Through the space–time Gibbs correspondence, Theorem A also gives an equilibrium statement with finite energies, Corollary B in Section 9: one computably specified translation-invariant finite-range interaction on $\mathbb{Z}^7$, with a finite alphabet and energies in $\{0, 1\}$, whose Gibbs measure invariant under all translations of $\mathbb{Z}^7$ is unique, on computable disjoint rational open intervals $\mathcal{J}_n$ of inverse temperature, exactly when $U(n)$ halts. This answers [10, Open problem 9.1] in dimension seven: take β_n to be the midpoint of $\mathcal{J}_n$.

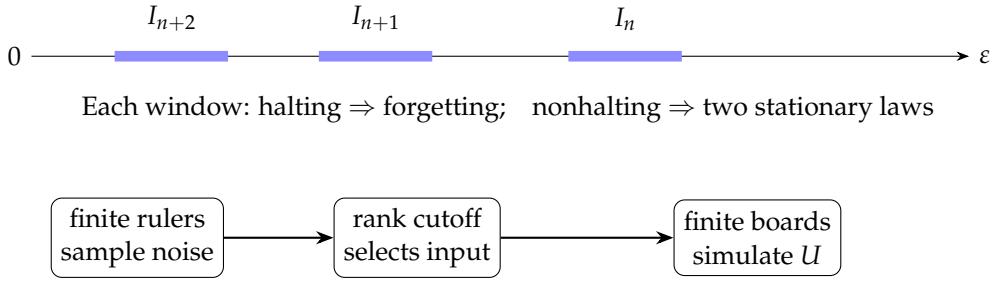


Figure 1: The intervals and their labels are schematic. The interval endpoints are computable; their ergodicity labels encode halting. The local rule uses finite sensor fields and the chart's finite geometry.

Relation to earlier work. Toom's stability theorem and Gács's reliable automata established memory under positive noise [28, 13]. Undecidability of ergodicity across descriptions goes back to Kurdyumov [20]; see [31, Chapter 14]. Toom proved it for binary PCA with transition probabilities in $\{0, \frac{1}{2}, 1\}$ and for continuous-time particle systems [29, 30]. Bušić, Mairesse and Marcovici showed that a deterministic one-dimensional CA is ergodic exactly when it is nilpotent [3, Theorem 3.3]; Kari proved nilpotency undecidable [19]. Marcovici, Sablik and Taati developed ergodicity criteria for noisy classes [21]. Marsan, Sablik and Törmä

Result	Object held fixed	Parameter or input
Bušić–Mairesse–Marcovici [3]	Description varies	Undecidable ergodicity, already for deterministic CA
Marsan–Sablik–Törmä [25]	One CA on $\mathbb{Z}$	At least two noise transitions
Marsan–Sablik [24]	One continuous Cantor map; shift commutation not required	Every G_δ set containing 1, as uniqueness and ergodicity set; every Π_2-computable one as the uniqueness set of a computable map (their Section 5)
Forgetting [8]	Computable family on $\mathbb{Z}^8$	Halting selects ergodicity at fixed noise
Windows [9]	One CA on $\mathbb{Z}^4$	Infinitely many alternating computable windows
Hard-constraint [10]	One classical hard-constraint model	Halting-labelled Gibbs uniqueness windows
Theorem A	One CA on $\mathbb{Z}^6$	Halting-labelled uniform-noise windows

Table 1: The closest results and the parameter in which their conclusions vary.

constructed a fixed one-dimensional automaton with at least two ergodicity transitions and asked which sets of rates can occur [25, Section 6]: "which sets can be obtained as" $\{\varepsilon \in [0,1] : F_\varepsilon \text{ is ergodic}\}$. For continuous Cantor maps under uniform noise, Marsan and Sablik realize every G_δ subset of $[0,1]$ containing 1 simultaneously as the uniqueness set and the ergodicity set [24, Theorem 3.19]. They state that the uniqueness set of a computable map is Π_2-computable in their sense, that every Π_2-computable G_δ subset of $[0,1]$ containing 1 is the uniqueness set of a computable map, and that for cellular automata it is still open to characterize the possible uniqueness sets [24, Section 5]. Their earlier work realizes sets of zero-noise limiting measures, using computations whose time budget is set by the noise rate [23, Sections 5.1–5.2.1]. Marcovici and Taati prove exponential forgetting for positive-rate PCA admitting a stationary Bernoulli law [22]. The distinction between a family of rules and one fixed rule is recorded in Table 1.

Hochman and Meyerovitch characterized multidimensional SFT entropies [17]. Chazottes and Hochman, and Gayral, Sablik and Taati studied zero-temperature Gibbs limits and their computability [4, 15]. Cubitt, Perez-Garcia and Wolf proved spectral-gap undecidability [6, 7]; Bausch, Cubitt and Watson gave a continuous quantum parameter family with an uncomputable phase diagram [1], and Purcell, Li and Cubitt placed a transition at an uncomputable number [27].

What is ours. Theorem A realizes halting-labelled windows for a cellular automaton: a shift-commuting map with one finite neighbourhood, within the class whose possible diagrams are left open in [24, Section 5]. Our construction combines a fixed computational chart, a finite-range interval controller, a joint estimate for sensor feedback, and a common-noise coupling. The hierarchical scaffold is from our hard-constraint paper [10]. Our forgetting paper supplies the checker/quorum approach [8]; our windows paper asks the fixed-rule question answered here [9]. Our nilpotent paper gives, for each n, a nilpotent rule with n separated memory windows and the mechanism of independent gates read from diagonal copies [11, Section 3, Exact gate law]; here the intrinsic gate sources follow diagonal characteristics

as well. Corollary B uses the correspondence between translation-invariant stationary laws and translation-invariant space–time Gibbs measures due to Goldstein, Kuik, Lebowitz and Maes [16], in the form proved in our space–time paper [12, Lemma 2.4 and Corollary 10.4]. The proof below establishes the stochastic statements needed for this construction and proves finite-board forcing in every phase.

Dimension and scale. Two coordinates carry the chart and four carry the threshold checker. The interval endpoints are specified by terminating rational searches; Section 9 gives their scale.

Reading guide. Section 9 ends with the Gibbs corollary on $\mathbb{Z}^7$. Section 2 gives the noisy dynamics and its complete local rule, with constants selected in Section 9. Sections 3 and 4 construct the chart and prove the all-phase board lemma. Sections 5 and 6 give the contour and sensor estimates. Section 7 proves memory; Section 8 proves forgetting from every initial state. The imported scaffold facts are isolated in Proposition 3.1. Readers interested in the stochastic mechanism can take that proposition as the cited input and read Sections 2 and 5 to 8.

2 Dynamics and the fixed local rule

A cellular automaton on a finite alphabet Σ and lattice $\mathbb{Z}^d$ is a map F given by one finite neighbourhood N_F and one function $f : \Sigma^{N_F} \to \Sigma$. Its uniform replacement perturbation has independent variables $E_t(w) \sim \text{Bernoulli}(\varepsilon)$ and $Z_t(w)$ uniform on Σ, at all sites and integer times. The update is

$$X_t(w) = \begin{cases} Z_t(w), & E_t(w) = 1, \\ F(X_{t-1})(w), & E_t(w) = 0. \end{cases}$$

The replacement is of the entire product symbol. All coupled trajectories use these same variables. A stationary law is fixed by this Markov kernel. Ergodicity means a unique stationary law and weak convergence to it from every initial law. Translation invariance means invariance under every spatial translation.

2.1 The planar checker and provider threshold

For a finite Wang set R, let $\mathcal{L}_R$ consist of pairs (c, b), where c is an edge-matching R-pattern on $\{-1, 0, 1\}^2$ and $b \in \{0, 1\}$. These patterns are enumerated by finite local checks. Two letters at displacement $\zeta \in \{-1, 0, 1\}^2 \setminus \{0\}$ are compatible when their R-blocks agree on the overlap after translation by ζ and their bits agree. Put $\mathcal{A}_R = \{\perp\} \cup \mathcal{L}_R$. A letter other than $\perp$ is live. A full legal chart means an assignment of live letters compatible in all eight Moore directions. Equivalently it is the 3×3 block presentation of an R-tiling with one global bit. The controller reads the central projected tile of a live letter.

Write a site as $(u, P) \in \mathbb{Z}^2 \times \mathbb{Z}^4$. Fix

$$m = 16, \quad L = 160, \quad K = 16^4, \quad q = 5K, \quad h = 4K. \tag{1}$$

Let $C_0 = 0$, $C_i = L\mathbf{e}_i$ for $1 \le i \le 4$, $S_i = C_i + \{0, \dots, 15\}^4$ and $S = \bigcup_{i=0}^{4} S_i$. For a data configuration D_0, check each provider $(u, P + s)$ by retaining its central live letter exactly when its eight Moore neighbours are live and compatible with that centre at their respective

displacements. A failed check returns $\perp$. Define $F_R(D_0)(u, P)$ as a live letter occurring in at least h checked providers, and $\perp$ if none does. Since $2h > q$, that letter is unique. A legal chart cloned over P is fixed by F_R.

2.2 Interval fields

Set $T = 500$ and $v = (32, 32, 32, 32)$. The remaining fixed constants C, $B = 4^k$, s, and n_0 are selected in Section 9. In particular $s = \lceil 3CB/4 \rceil$. The helper alphabet $\mathcal{H}$ has 100 letters: LOW represents $\{0\}$, UNKNOWN represents $\{0, 1\}$, and 98 HIGH aliases represent $\{1\}$. Fix one representative for each of these three intervals.

The physical alphabet is

$$\Sigma = \mathcal{A}_R \times \mathcal{A}_R \times \mathcal{H}^B. \tag{2}$$

Its coordinates are data D, a second coordinate H, and fields $I_0, \dots, I_{B-1}$. The second coordinate records the ordinary proposal and is retained in the physical symbol. All drivers read old data. At proposal time t put

$$A_t = F_R(D_{t-1}), \qquad Q_t(u, P) = F_R^T(D_{t-1})(u, P - (T-1)v). \tag{3}$$

The chart alphabet provides predicates Orig, End, a row flag H_{row}, and a lane phase $p \in \mathbb{Z}/B\mathbb{Z}$, defined in Section 3. Let $\mathbf{e}_x, \mathbf{e}_y$ denote the planar unit vectors; a provider vector in a site expression changes only P.

For a physical site w, field j always uses the base $z = w - j\mathbf{e}_y$. Its provisional interval is

$$J_{j,t}(w) = \begin{cases} \{0,1\}, & A_t(z) = \perp, \\ \{0\} \cup I_{j,t-1}(w - \mathbf{e}_x + v), & A_t(z) \neq \perp,\ Q_t(z) = \perp, \\ \{0\}, & A_t(z), Q_t(z) \neq \perp,\ \mathrm{Orig}(Q_t(z)), \\ I_{j,t-1}(w - \mathbf{e}_x + v), & \text{otherwise.} \end{cases} \tag{4}$$

Intervals on the right are interpreted through their represented subsets. At a base w, let h_w count the provisional intervals $J_{j,t}(w + j\mathbf{e}_y)$ containing 1, and l_w those equal to $\{1\}$. All of them use the same $A_t(w), Q_t(w)$. Define

$$\begin{aligned} \mathrm{Pin}_t(w) = 1 \quad \Longleftrightarrow \quad & Q_t(w) = \perp \quad \text{or} \\ & \big(Q_t(w) \neq \perp,\ p(Q_t(w)) = 0,\ \mathrm{End}(Q_t(w)),\ \text{and} \\ & \quad \big[(H_{\mathrm{row}}(Q_t(w)) = 1\ \&\ h_w \geq s)\ \text{or} \\ & \quad\ (H_{\mathrm{row}}(Q_t(w)) = 0\ \&\ l_w < s)\big]\big). \end{aligned} \tag{5}$$

The deterministic proposed symbol is

$$\left(\begin{cases} \perp, & \mathrm{Pin}_t(w) = 1, \\ A_t(w), & \mathrm{Pin}_t(w) = 0, \end{cases} \quad A_t(w), \quad (\mathrm{rep}\, J_{j,t}(w))_{j<B} \right). \tag{6}$$

One uniform whole-symbol replacement follows. These cases also specify the rule on malformed configurations and on every exceptional wall: the predicates are local tile functions. Every offset and every iteration depth in (3)–(6) is fixed.

Proposition 2.1. *Once R, B, s are fixed, (6) defines one computably specified finite-range cellular automaton. The number of parent sites in both dependency directions together is at most*

$$\kappa = 2B(n_A + n_Q + 2), \qquad n_A = 9q, \qquad n_Q = (2T+1)^2[T(L+m-1)+1]^4.$$

This bound is independent of $|R|$ and n_0.

Proof. An ordinary query reads at most $9q$ old data sites. In T iterations the planar displacement lies in $[-T, T]^2$ and each provider coordinate in $[0, T(L+m-1)]$; recentering translates this box. At output w the A, Q bases are the B sites $w - j\mathbf{e}_y$, including w. The gate's provisional fields use the already counted base w and B predecessor helper sites. The helper outputs add one predecessor site. Thus there are at most $B(n_A + n_Q) + (B+1) \le B(n_A + n_Q + 2)$ parents. Translation invariance gives the same reverse degree. Alphabet enlargement changes finite lookup tables, with the checker radius remaining one. □

3 Charts, frames and routing

3.1 The cited scaffold

We use the scaffold of [10, Section 3 and Appendix A], with the base alphabet, edge endpoints and substitution tables specified there. Its base substitution is Ollinger's [26], with the ruler decoration from [8]. Its 276-letter decorated Wang set is denoted $\mathbb{A}$. For a physical coordinate $z = \delta_i + 2u + \sigma$, $\sigma \in \{0,1\}$, parent rank means $\nu_2(u - \varpi_i)$, with rank infinity when $u = \varpi_i$. Its thin, thick and black components, the ruler bits χ_1, χ_2, and its quarter alphabet have the meanings of those tables. The following is the precise imported input.

Proposition 3.1 (Scaffold facts)**.** *The scaffold has an injective expansion-two substitution, preserves legality in both directions, and every tiling has a unique desubstitution tower. Its quarter tilings have a unique quarter offset $\delta \in \{0,1\}^2$ and phase $\tau = \delta + 2\varpi \in \mathbb{Z}_2^2$. Every phase occurs; its residue modulo 2^l is a local function of the tiling. For Haar-almost every phase the tiling is determined by the phase, every coordinate has finite parent rank and belongs to a finite frame.*

On axis i put $a_d^{(i)} = \tau_i \bmod 4^{d+1}$ in $[0, 4^{d+1})$. The depth-d frames are

$$I_d^{(i)}(z) = [a_d^{(i)} + (4z+1)4^d + 1,\ a_d^{(i)} + (4z+3)4^d]. \tag{7}$$

Same-depth frames are disjoint; frames of different depths are disjoint or strictly nested. Every deeper frame contains a frame of each smaller positive depth. The free coordinates $\mathcal{F}_d^{(i)}(z)$ are the interior coordinates outside all smaller-depth frames. They number $N_d = 2^{d+2} - 2$, and their extremes are the frame endpoints plus one and minus one.

A column of finite parent rank r carries light vertical wall sites only if $r = 2e - 1$ and it is an endpoint of a depth-e column frame; it then carries them exactly at the rows in depth-e row frames, as a left or right side according as it is a left or right endpoint. The horizontal statement follows by exchanging axes. A depth-e endpoint has parent rank $2e - 1$ and ruler bit 1. More generally χ_i equals parent-rank parity at finite rank.

The last paragraph uses the Ruler bits and Red paths lemmas [10, Lemmas A.6 and A.7]. Their finite-rank statements hold for all tilings. In particular the column statement assumes only the column's finite rank. The global light-wall statement in that paper assumes finite ranks on both axes; below we apply the finite-rank statements within one closed frame.

At a finite-rank row/column intersection the black type is A for equal ranks, B or C if the column rank is smaller, and D or E if it is larger. Indeed, the substitution table [10, Table 4] makes child $(0,0)$ keep the black type, child $(1,1)$ have black A, child $(1,0)$ have black in $\{\mathsf{B}, \mathsf{C}\}$ and child $(0,1)$ in $\{\mathsf{D}, \mathsf{E}\}$. The letter descends through children $(0,0)$ from its ancestor at the first nonzero digit pair of the parent coordinate $u - \varpi$; that pair is $(1,1)$, $(1,0)$ or $(0,1)$ according as the column rank equals, is smaller than, or is larger than the row rank.

3.2 Rank flags and lane phase

Add a column bit V and a row bit H_{row} before quartering. Copy V on north/south edge colours and H_{row} on east/west colours. Impose

$$\mathsf{A}:\ V = H_{\mathrm{row}}; \qquad \mathsf{B},\mathsf{C}:\ V \geq H_{\mathrm{row}}; \qquad \mathsf{D},\mathsf{E}:\ V \leq H_{\mathrm{row}}. \tag{8}$$

Internal quarter edges compare the whole decorated parent; external edges compare its edge colours. This is a finite Wang construction.

Lemma 3.2. *In any legal decorated chart the flags at finite ranks have one common nonincreasing value sequence $(b_r)_{r\geq 0}$. Every finite cutoff $b_r = \mathbf{1}\{r < c\}$ occurs over every typical scaffold chart.*

Proof. Rows and columns carry constant bits by edge matching. Equal-rank intersections have black type A, so all rank-r rows and columns share one value b_r. Intersecting rank $r < s$ in the two possible orders and applying (8) gives $b_r \geq b_s$. Conversely assign the displayed cutoff to each row and column. The three black-type rank comparisons just proved verify every local filter. All ranks are finite in a typical chart. □

The lane phase is

$$p(x,y) = y - \tau_2 - 1 \pmod{B}. \tag{9}$$

It is implemented by a higher-block Wang extension. Here is an effective construction of its lookup table. For residues modulo B encode the quarter offset and $2k-1$ base substitution digits. Enumerate finite hierarchy patches of increasing radius, with the source's explicit substitution and matching constraints. Search for a central radius whose projected fine words for distinct residues are disjoint, and a finite hierarchy radius witnessing this disjointness. Each test is finite exhaustive enumeration. The search terminates: phase sectors are disjoint compact sets by the tower and local phase facts of Proposition 3.1; if every larger hierarchy patch allowed an ambiguous fixed central word, compactness would give ambiguity in two full sectors. There are finitely many central words and sector pairs. Decode a liftable word by its unique residue and give an unliftable word a fixed default. Require p to agree with that table, to copy horizontally and increase by one vertically. The usual overlapping-block edge colours recode these finite constraints as a Wang set, retaining the central tile projections.

Define Orig as the intersection of a bottom light horizontal wall with a left light vertical wall, and End with a right light vertical wall. These are literal functions of a quarter tile's red atoms, quadrant and ruler bits from [10, Definition 3.11]. On a finite bottom row of depth d, origins and ends are respectively

$$(L_d(z), \mathrm{Bot}_d(z')), \quad (R_d(z), \mathrm{Bot}_d(z')), \qquad \ell_d = R_d - L_d = 2\cdot 4^d - 1.$$

Here Bot_d denotes the lower endpoint of a row frame. Finite-rank Red paths gives this assertion even if the other phase coordinate is exceptional. At this row

$$p(\mathrm{Bot}_d(z')) \equiv (4z'+1)4^d \pmod{4^{\min(d+1,k)}},$$

so the phase-zero gates are exactly the finite bottom rows with $d \geq k$.

3.3 Local routing rules

Retain the orientation/event fields of [10, Section 4, (O1)–(E3)]. We give their constraints here. At each side there is an orientation pair: entries in $\{\mathrm{L},\mathrm{R}\}$ on horizontal edges, in $\{\mathrm{B},\mathrm{T}\}$ on

vertical edges. There is also an event tag in $\{\text{cell}, \text{wall}\}^2 \cup \{*\}$. Matching edges identify the entire tags. Let $\mathrm{VW} \in \{-, \mathrm{L}, \mathrm{R}\}$ and $\mathrm{HW} \in \{-, \mathrm{B}, \mathrm{T}\}$ be the local wall predicates.

If $\mathrm{VW} = -$, the west and east orientation pairs agree. At a wall of side a, the plus entry on its west edge and minus entry on its east edge both equal a. Apply the transposed rule vertically. An event tag is a pair precisely when its orientation is (L, R) horizontally or (B, T) vertically, and is $*$ otherwise.

A site is a horizontal carrier when $\mathrm{VW} = -$ and the west orientation is (L, R); define vertical carriers by transposition. An active site is both. A horizontal-only site copies both horizontal event entries; an active site puts cell in its west plus and east minus entries. At an inward-facing left wall put wall in the east minus entry, and at an inward-facing right wall put wall in the west plus entry. The vertical rule puts cell in the south plus and north minus entries of active sites, copies at vertical-only sites, and puts wall at inward-facing bottom/top walls. These are (E2)/(E3).

On carriers introduce horizontal logical colours in $\{\circ\} \cup S_P$ and vertical ones in $\{\circ\} \cup S_P^2$, where S_P is the machine-content alphabet of Section 4. Nonhorizontal carriers have empty horizontal colours; horizontal-only sites copy those colours. Impose the transposed vertical conditions. Records are empty off active sites; at active sites they are genuine triples (t_S, t_C, t_N) with $t_C \in S_P$, $t_S, t_N \in S_P \cup \{\perp\}$. Define the four flags at an active site by

$$f_W = [e_-^W = \text{wall}], \quad f_E = [e_+^E = \text{wall}], \quad f_S = [e_-^S = \text{wall}], \quad f_N = [e_+^N = \text{wall}].$$

Require $\ell^X = \circ$ exactly when $f_X = 1$, for every side X. The remaining constraints are

$$\begin{aligned}
&t_S = \perp \ (f_S = 1), \quad \ell^S = (t_S, t_C) \ (f_S = 0),\\
&t_N = \perp \ (f_N = 1), \quad \ell^N = (t_C, t_N) \ (f_N = 0),\\
&t_C = \mathrm{init}(H_{\mathrm{row}}, \chi_2, f_S) \ (f_W = 1), \quad t_C = \ell^W \ (f_W = 0),\\
&\ell^E = f_P(t_S, t_C, t_N) \ (f_E = 0), \qquad t_C \text{ does not carry } \mathsf{HALT}.
\end{aligned} \tag{10}$$

The read-only input tracks in t_C agree with the local H_{row}, χ_2. This retains precisely (K1)–(K3), the boundary rule (C0), and the displayed replacement for (C5). The finite alphabet R is the set of tuples passing these tests, followed by the phase extension above. Each tuple test and each colour equality is effective.

3.4 Forcing a finite board in every phase

Lemma 3.3 (Finite-board forcing)**.** *In any scaffold phase, every extension satisfying the preceding rules contains, for each $d \geq 1$ and each pair of depth-d frames $[L_*, R_*]$, $[\mathrm{Bot}_*, \mathrm{Top}_*]$, the board*

$$\{(x_i, y_j) : 0 \leq i, j < N_d\} = \mathcal{F}_d^{(1)} \times \mathcal{F}_d^{(2)}.$$

Here $x_0 < \cdots < x_{N_d-1}$ and $y_0 < \cdots < y_{N_d-1}$ list the free coordinates. Its consecutive horizontal and vertical channels are single carriers. Its boundary flags are exactly

$$f_W = [i = 0], \quad f_E = [i = N_d - 1], \quad f_S = [j = 0], \quad f_N = [j = N_d - 1].$$

Its central contents $s_i(j) = t_C(x_i, y_j)$ obey

$$\begin{aligned}
s_0(j) &= \mathrm{init}(H_{\mathrm{row}}(y_j), \chi_2(y_j), [j = 0]),\\
s_{i+1}(j) &= f_P(s_i(j-1), s_i(j), s_i(j+1)), \quad i < N_d - 1,\\
s_i(-1) &= s_i(N_d) = \perp.
\end{aligned}$$

Thus it contains the unique N_d-cell, $(N_d - 1)$-step finite-tape computation.

Proof. Write $z - \delta_i = 2u + \sigma$ for a physical coordinate and its quarter digit $\sigma \in \{0, 1\}$. Its parent rank is $\nu_2(u - \varpi_i)$. If that rank is at least $2d + 1$, including infinity, then

$$z - \tau_i = 2(u - \varpi_i) + \sigma \equiv \sigma \pmod{4^{d+1}}.$$

Every coordinate in the closed frame has residue in $[4^d + 1, 3 \cdot 4^d]$, which contains neither 0 nor 1. All coordinates of the closed rectangle therefore have finite parent rank at most $2d$. Equivalently the rectangle sits strictly inside one quartered level-$(2d + 1)$ supertile. We may apply the finite-rank wall classification at every site of its walls and channels.

Fix a free row y_j. The only vertical light walls on $[L_*, R_*] \times \{y_j\}$ are its two endpoints. Shallower walls cannot occur because y_j lies in no shallower row frame. Same-depth column frames are disjoint. A deeper endpoint cannot lie in the shallower frame by strict nesting. Orientation matching across the open strip is constant; the left wall fixes its minus entry to L and the right wall its plus entry to R. The strip is therefore horizontally carried with orientation (L, R).

We must determine its vertical carriers as well. For $z \in (L_*, R_*)$ let

$$D_z = \{e : 1 \le e \le d,\ z \text{ lies in a depth-}e \text{ column frame}\}, \qquad e_z = \min D_z.$$

This set contains d. Consider all depth-e row frames with $e \in D_z$ meeting $[\mathrm{Bot}_*, \mathrm{Top}_*]$. They form a finite strictly laminar family containing $[\mathrm{Bot}_*, \mathrm{Top}_*]$. By the wall classification their endpoints are exactly the horizontal walls on this strip. Its minimal members are exactly its depth-e_z members: every deeper member contains a depth-e_z frame, and none of smaller eligible depth exists. Consecutive endpoints labelled B then T bound exactly a minimal member. Indeed, if the interval starting at the earlier endpoint ended later than the second, the interval ending at the second would have to start earlier than the first, contradicting disjointness or strict nesting. The converse follows because a minimal interval contains no endpoint in its interior. The vertical orientation rule forces each gap's pair from its two endpoint labels. Hence vertical carriers are exactly the interiors of those minimal frames.

If z is free at depth d, then $e_z = d$ and every (z, y_j) is a vertical carrier. If z is not free, then $e_z < d$, while y_j lies in no depth-e_z frame, so (z, y_j) is horizontal only. This proves that the active sites on each free row are exactly the x_i, with horizontal-only interiors between consecutive ones. Transposing proves the vertical-only assertion. The extreme free coordinates are $L_* + 1, R_* - 1, \mathrm{Bot}_* + 1, \mathrm{Top}_* - 1$ by Proposition 3.1; their adjacent exterior sites are the four light walls.

All horizontal event tags in this strip are pairs. The left wall sets the inward minus entry to wall; the right wall sets the inward plus entry to wall. Each active site supplies the facing cell entries, and the single-carrier channels copy both entries. The successive gap pairs, including gaps consisting of a single shared edge, are

$$(\text{wall}, \text{cell}), \qquad (\text{cell}, \text{cell}), \qquad (\text{cell}, \text{wall}).$$

Thus the west/east flags are exactly the two extreme-index indicators. The transposed event argument proves the south/north flags.

Horizontal-only sites transmit horizontal colours and have empty vertical colours and records. The left/right walls are not horizontal carriers, so their horizontal colours are empty. Consequently consecutive logical cells share one horizontal colour and the two exterior faces are empty. Apply the same argument to vertical channels and bottom/top walls. The boundary rule and the exact flags now force every internal logical face to be nonempty and

every outer one empty. Every active site has a triple record. Exterior choices cannot disconnect or inject records into this network.

The shared vertical colour at (i, j) and $(i, j+1)$ has descriptions

$$(s_i(j), t_N(i, j)) = (t_S(i, j+1), s_i(j+1)).$$

Thus $t_N(i, j) = s_i(j+1)$ and $t_S(i, j+1) = s_i(j)$, with bottom symbols on the outer boundaries. At $i = 0$ the west flag initializes the contents. At $i < N_d - 1$ the east colour is $f_P(t_S, t_C, t_N)$ and is passed to the next cell's central content. Substitution gives the stated recursion. It determines each time column, then all remaining record entries and colours. No successor is demanded on the last east boundary. The machine conversion in Lemma 4.1 identifies this recursion with the one-head computation and sends attempted exits to FAIL. □

4 The computation carried by a chart

Take U to be a fixed deterministic one-way-tape universal machine on positive unary inputs, with finite tape alphabet Γ_U containing blank and 1. Its finite transition table is part of the specification. Absorbing states FAIL, HALT are distinct from its working states. The integer n_0 is fixed by Section 9.

4.1 A direct finite machine

The machine P has read-only bits h_{in}, π, z, writable bits p_*, e_*, a writable track $n_* \in \{0, 1, 2\}$ and a writable track $u_* \in \Gamma_U$. Initially $h_{\text{in}} = H_{\text{row}}$, $\pi = \chi_2$, $z = [j = 0]$, the work tracks are zero or blank, and the sole head is at cell zero in SCAN. A transition preserves every track not explicitly written. All unspecified cases and every attempted tape exit go to absorbing FAIL. Right means increasing logical tape index j (north). The following clauses are a finite transition prescription. A return to the origin with continuation Z means state RET_Z, moving left while $z = 0$ and staying with state Z at $z = 1$.

1. In SCAN, on $h_{\text{in}} = 1$ write $p_* = 1$ and move right in SCAN. On $h_{\text{in}} = 0, \pi = 0$ fail. On $h_{\text{in}} = 0, \pi = 1$ write $p_* = 1$, move right in PLUS. PLUS writes $p_* = 1$, moves right in SETEND. SETEND writes $e_* = 1$, moves left, and returns with continuation R_{00}.

2. States R_{bc} have $b \in \{0, 1\}$, $c \in \{0, 1, 2\}$. At $e_* = 0$, a zero p_* is skipped. A one p_* is retained if $b = 0$ and erased if $b = 1$; move right, flip b and replace c by $\min(c + 1, 2)$. At $e_* = 1$ apply these clauses in order: $c = 0$ fails; $c = 1$ returns with continuation SKIP_0; $c = 2, b = 1$ fails; $c = 2, b = 0$ returns with continuation APPENDN.

3. APPENDN scans right over $n_* = 1$. At the initial zero it writes one and returns with continuation R_{00}; a two fails. Thus every even halving appends one token.

4. For $0 \le c < n_0 + 1$, SKIP_c requires $n_* = 1$, changes it to two, and moves right to SKIP_{c+1}. State SKIP_{n_0+1} requires another one and returns with continuation FINDN.

5. FINDN scans over twos. On a one it writes two and returns with continuation APPENDU. On a zero it returns with continuation $\text{RUN}_{q_{U,0}}$. APPENDU scans right over $u_* = 1$, writes one at the initial blank and returns with continuation FINDN; every other tape symbol fails.

6. RUN_q executes the literal transition table of U on u_*, preserving the other tracks. A halting successor, or an initially halting state, enters HALT. Both HALT and FAIL stay forever.

Let S_P be the resulting finite alphabet of tape symbols with one state or a no-head mark. Define $f_P(a, b, c)$ by this ordered rule: if b carries a head, perform its transition, retaining a failing head if the move points to a $\perp$ neighbour (a tape end); otherwise retain its new head for a stay move or drop it for a move. If b has no head but a moves a head north, retain b's symbol and install that new head state. Otherwise do the same for a head from c moving south. In the remaining case preserve b without a head. This is total even on multiple-head triples, with the stated priority.

Lemma 4.1. *On a finite tape initialized above, f_P gives exactly the sequential P computation, with one head at every time and absorbing failure on an attempted exit. If the initial zero of h_{in} has index j_*, P can begin simulating U only if its parity bit is one and $j_* + 2 = 2^e$ with $e > n_0 + 1$. It then runs exactly $U(e - n_0 - 1)$.*

Proof. A stay step changes just the head cell. For an interior move the old head cell loses the head and exactly its destination receives the new state, retaining the destination's own tape symbol. Every other cell preserves its symbol. An exit leaves the head at the boundary in FAIL. Induction gives the one-head assertion and excludes the malformed priorities.

The scan makes exactly $j_* + 2$ marks, with e_* on the next cell. Each R pass keeps the odd-numbered surviving marks. Its c records their original count capped at two, and b its parity. The ordered termination clauses accept a singleton before rejecting an odd number exceeding one. Consequently an original power 2^e has exactly e successful even passes; any other positive count eventually fails. APPENDN therefore writes e tokens. SKIP consumes $n_0 + 1$ of them and requires a further token. FINDN and APPENDU copy precisely the remaining $e - n_0 - 1$ tokens to the initially blank u_* track. Only RUN can enter HALT, and it executes U on that unary input. Any shortage of tape during any routine fails. □

4.2 Which input is read

Lemma 4.2. *Suppose $a \geq 2$ and the finite-rank flags satisfy $b_{2a-3} = 1$, $b_{2a-1} = 0$. Every depth-d board with $d > a$ has its initial zero at logical index $2^{a+1} - 2$, with odd parent rank $2a - 1$. A board with $d \leq a$ either has no zero before its boundary or has a zero with even rank before any odd-rank zero. Thus it cannot simulate a different input.*

Proof. By Lemma 3.2 the initial zero rank is either $2a - 2$ or $2a - 1$. In a depth-d frame of phase τ_i, write a free coordinate as $a_d^{(i)} + (4z + 1)4^d + t$, starting with $t = 2$. In its left half, before $t = 4^d$, the parent rank is $\nu_2(\lfloor t/2 \rfloor)$. An even rank $2e$, $1 \leq e < d$, gives residues $2 \cdot 4^e$ or $2 \cdot 4^e + 1$ inside a smaller excluded frame. Hence noncentral free ranks are zero or odd. The two central coordinates have rank $2d$.

For $d > a$, the initial free coordinate of rank at least $2a - 2$ is $t = 4^a$, of rank $2a - 1$. To count its index let A_a be the residues in $[0, 4^a)$ excluding every smaller-frame window. There are four at $a = 1$. The four next-digit blocks contribute $|A_a|, 1, 0, |A_a| - 1$, so $|A_a| = 2^{a+1}$. Allowed coordinates in $[2, 4^a]$ remove zero and one and add 4^a, giving $2^{a+1} - 1$ coordinates. The initial zero index is one less.

If $d < a - 1$, all board ranks are below the cutoff. If $d = a - 1$ an initial zero can occur only at the even central rank $2a - 2$; if $d = a$ it again occurs at the even centre. The parity guard rejects these cases. This uses only relative finite residues and holds in every phase. □

Proposition 4.3 (The computational chart)**.** *For $a = n + n_0$, if $U(n)$ does not halt there is a spatially translation-invariant law of legal charts with either global bit and with row flag one on finite*

bottom depths $d < a$ and zero on depths $d \geq a$. If $U(n)$ halts, every legal chart in every phase has a depth-$(a-1)$ bottom end with flag zero or a depth-a bottom end with flag one.

Proof. For nonhalting choose a typical phase and cutoff $2a-2$ or $2a-1$. In a typical phase, [10, Propositions 5.1 and 5.7 and Lemma 5.3] give a legal assignment of the orientation and event tags, with active sites exactly the grids $\mathcal{F}_d^{(1)} \times \mathcal{F}_d^{(2)}$, each active site in one grid, and the boundary flags of Lemma 3.3. These assignments use only (O1)–(E3), which we retain. Fill every board with its initialized finite run. Large boards decode n by Lemmas 4.1 and 4.2; small boards fail or reach a boundary. A finite exit fails, so none contains HALT. Fill channel colours from their board and put empty records elsewhere. These assignments satisfy all tuple filters, and the phase extension is unique. Both global bits are compatible. Translating and taking a weak limit of spatial averages gives an invariant law. Its phase is Haar because each finite residue law is invariant under the transitive integer translations of that residue torus. Hence it gives full measure to typical phases and retains the cutoff.

For halting, avoiding both asserted wrong flags would impose the two rank conditions of Lemma 4.2, since every finite depth has bottom corners in every phase. Every $d > a$ board would decode n. The decoding and the halting computation have finite time and space budgets. Choose d with N_d larger than the elapsed time and largest visited address. Lemma 3.3 forces its computation, including HALT at a stored time column, contradicting (10). Both critical depths are gated since $a - 1 \geq k$. □

5 Contour estimates for the data checker

5.1 Finite supports and charge arms

The charge-arm form of Toom's contour method is described by Gács [14], building on the proof of Berman and Simon [2]. We prove the finite estimate used here. A two-cluster support consists of one selected parent in each of two distinct provider clusters, with a Moore planar displacement allowed. For each charge it provides a good parent as follows: charge zero chooses a nonzero cluster; charge $i \geq 1$ chooses a cluster other than i. For backward physical steps define

$$\psi_0(P,t) = 5\sum_i P_i + 640t, \qquad \psi_i(P,t) = -5P_i - 160t \quad (1 \leq i \leq 4).$$

Their sum is zero. On $P \mapsto P + s$, $t \mapsto t - 1$ the good gains are at least 160 for charge zero and

$$\varrho_+ = 160 - 5 \cdot 15 = 85 \tag{11}$$

for the others. Every gain is at least $-\varrho_-$, where $\varrho_- = 4 \cdot 160 + 5 \cdot 15 = 715$. Put

$$A_0 = 5(1 + \varrho_-/\varrho_+) = 4000/85 < 48.$$

Planar displacements do not affect these potentials.

Lemma 5.1 (Grouped terminal estimate)**.** *In a finite root-reachable acyclic support graph, let every nonterminal have two selected supports and a good support for each of the five charges. Partition its terminal leaves into groups. Suppose the sum of the five potential oscillations within any group is at most Ω. One can select g distinct groups and a connected copied-arm witness containing the root with*

$$e \leq A_0(g-1) + \Omega g/\varrho_+ \tag{12}$$

positive support edges, counted with multiplicity. Unary zero-potential links are permitted; they create no additional charge sources. With singleton groups a nonterminal root has at least two terminal flags. If support edges decrease time by one and group identifications preserve time, the span is at most e.

Proof. Give each vertex its set of descendant terminal-group names. Expose the graph in topological order, replacing a frontier nonterminal by its supports. Join frontier vertices when their descendant sets overlap. Replacing a binary vertex splits its component into at most two: each remaining vertex remains linked to one support by the old shortest overlap chain. It joins no outside component because support descendant sets are subsets of the removed set. A unary replacement cannot split a component. At the end each component is one terminal group.

A copied-arm witness is the graph obtained by taking separate copies of the arm interiors, identifying their sources and identifying terminal leaves in the same group. Start five good arms from the root. Maintain exactly one arm endpoint of each charge in each frontier overlap component. When a component splits, add a source at the split vertex. For each charge choose its initial support in the new component lacking that charge endpoint, then follow good supports. The split leaves each charge's old endpoint in one of the two components, and the new source supplies its missing endpoint in the other; a unary exposure moves an endpoint within its component. Only this initial edge of a nonroot source can be bad. Copy arm interiors, merge at sources and merge leaves by group name, and take the connected component containing the root. For each charge its sources and terminal groups are in bijection. For each charge, following its arms pairs every source with exactly one terminal group, and every selected group has exactly one such endpoint; thus the root component has g sources and g groups, with one incidence of each charge at each.

At most $5(g-1)$ edges are bad. Summing charge increments, the source sums vanish. Comparing each group's charge endpoints to one reference leaf bounds the total terminal sum by $g\Omega$. Thus

$$\varrho_+ e - (\varrho_+ + \varrho_-)5(g-1) \le g\Omega,$$

which proves (12). For a nonterminal root $e > 0$, so singleton groups require $g \ge 2$. A minimum-time vertex is terminal because any arm interior continues earlier. A connected path from the root to it uses at most e support edges, while terminal identifications change time by zero. This proves the span assertion. The argument for the edge count used acyclicity and potential oscillations, so it also applies to groups at unequal times when their potentials agree. □

5.2 The raw marked set and coherence

Let $\mathcal{M}^{t_0}$ start empty at physical time t_0. Mark each replacement flag and each site having a marked Moore parent in two distinct provider clusters. Let $\mathcal{M}$ be its increasing limit as $t_0 \to -\infty$. Membership has a finite derivation. This is a set determined by flags alone.

Lemma 5.2. *For the threshold in Section 9, almost surely no infinite backward ordinary dependency path is contained in $\mathcal{M}$. The assertion holds simultaneously at every root.*

Proof. A marked point has a contour by Lemma 5.1, with f distinct flags, at most $50(f-1)$ arm edges and span at most that number. A terminal root has $f = 1, e = 0$. Along a marked path attach a witness at its current root and restart one step before its earliest terminal. Selected time slabs are disjoint, so their flags are distinct. At most $50(f-1)+1 \le 50f$ path steps are covered by a witness; together its support and path edges cost at most $101f$ edges. The graph

has fixed degree at most the Δ of Section 9. A doubled-edge traversal and a terminal subset encode at most K_{mem}^{f} choices by Lemma 9.1. Their probability is ε^{f}. A path of length n forces $f \geq n/50$, and the geometric tail tends to zero. Countability handles all roots. □

Lemma 5.3 (Coherence of unmarked live letters)**.** *In every complete trajectory of* (6)*, all unmarked live data or second-coordinate readings at a fixed planar site and time have the same letter over all provider positions. Readings at neighbouring planar sites are compatible in all eight Moore directions. If any unmarked live reading exists, one full legal, time-constant planar chart agrees with every such reading at every time.*

Proof. Off a replacement, data is an erasure of $F_R(D)$ and the second coordinate equals $F_R(D)$. An unmarked output has marked Moore parents in at most one cluster, hence at most K affected provider offsets. A live output has at most $K = q - h$ failed checked providers.

For two roots at one planar position, reduce their provider ℓ^1 separation by one coordinate step. Within each cluster pair offsets with the corresponding unit shift; there are $5K(1 - 1/16)$ injective pairs. For each root discard at most K failed supports and K marked-parent offsets; injectivity means each discard removes at most one candidate pair. At least

$$[5 - 4 - 5/16]K = 45056 > 0 \tag{13}$$

remain. Each remaining pair gives unmarked live data parents with the respective retained labels and a provider separation smaller by one. At zero separation data/second-coordinate agreement follows from the common proposal off flags; for mixed roots use one additional update before applying the same argument. Induction on the separation proves equal labels. For a Moore-neighbour pair at the same provider position, the two checked-provider sets overlap in at least $2h - q = 3K$ offsets. At a common offset one check certifies that its retained centre letter is compatible with the second retained centre letter. This is the zero-provider-span compatibility base, including all four diagonals. For different provider positions, the same injective pairing and discards give unmarked DATA parents at the two planar sites with smaller provider span. Induction therefore proves compatibility at every span. Using $\mathcal{M}^{t_0}$, the induction applies once elapsed time exceeds the span plus one. In a complete trajectory send t_0 to $-\infty$; unmarked parents remain unmarked at every finite stage.

Let V_t contain the planar sites with an unmarked live reading. Among h checked supports at least $h - K$ have no marked Moore parent. One supplies unmarked data at all nine planar sites at time $t - 1$, retaining the central label. Therefore

$$V_t + \{-1, 0, 1\}^2 \subseteq V_{t-1},$$

and labels persist backwards on overlaps. A nonempty V_t expands to cover every planar site in the union over earlier times. The ancient union $\bigcup_{s \leq t} V_s$ is therefore the whole plane. The nested domains and persistent labels define a full chart; each compatibility relation is checked at a common sufficiently early time. Future domains use those same labels. All statements are deterministic on the common flag set, so they apply also to noncausally selected complete trajectories. □

Lemma 5.4 (Deep wrong readings)**.** *Relative to the chart in Lemma 5.3, a wrong live depth-T query in* (3) *has a finite contour with at least eleven distinct raw flags and at most $50(f - 1)$ support edges. All its flags are no later than the queried input time. At lag $\ell \leq W$ from a current output time t, its earliest flag time t_0 satisfies $t - t_0 \leq W + 50(f - 1)$.*

Proof. Every wrong live actual reading is marked by Lemma 5.3. A wrong live forecast retains its letter at at least h center providers. At each of its T virtual levels these wrong centers occupy two distinct clusters; append their finite marked derivations. Give a virtual $F_R^j(D_s)(u,P)$ the provider characteristic coordinate $P+vs+jv$, and actual $D_s(u,P)$ the coordinate $P+vs$. Every virtual support has displacement $s'-v$, $s'\in S$, as does the final step to actual data; its potentials are the five spatial potentials $5\sum\chi_{P,i}, -5\chi_{P,i}$. The gains are (11) and -715. Recentring in (3) places the query root at the current characteristic coordinate.

Apply Lemma 5.1. Every root-to-flag arm crosses all T virtual levels, so $e \geq T = 500$, and $e \leq 50(f-1)$ gives $f \geq 11$. Virtual levels precede no actual input flag in physical time; their bookkeeping levels account for part of e. The remaining physical span is at most e, and adding lag ℓ gives the stated bound. □

6 Noise sensors and their joint source law

Fix a legal reference chart and drive ideal interval fields by its exact instructions. A true origin resets to zero; other sites copy. The replacement mark is uniform on $\mathcal{H}$ because (2) is a product, even though the fields at a physical site share their flag. The upper scalar mark is zero for LOW and one for UNKNOWN or HIGH; the lower scalar mark is one for HIGH and zero otherwise.

The true ride of a finite ruler is the segment of its base characteristic from the origin output to the output preceding its end. On a finite ruler of depth d, the provisional end reads exactly ℓ_d update atoms, from the origin output through the output preceding the end. The end's own replacement is later than its test. Thus the ideal upper success probability is

$$\bar{p}_d = .99[1-(1-\varepsilon)^{\ell_d}]. \tag{14}$$

For a recent window of W updates a lane bit is one exactly when its most recent replacement mark is a HIGH alias; it is zero for LOW, UNKNOWN, or no replacement. Its probability is

$$p_W = .98[1-(1-\varepsilon)^{W}]. \tag{15}$$

A short bad event means at least s upper successes. A long bad event means fewer than s successes in this recent-window experiment.

Lemma 6.1 (Ownership of noise atoms)**.** *For one fixed chart phase, the intrinsic source supports of distinct finite phase-zero gates are disjoint, over all space and time. Their B lanes are independent. The same assertion holds if a long support is restricted to its recent W atoms.*

Proof. A phase-zero row y owns the physical rows $y,\dots,y+B-1$, using field j at row $y+j$. These blocks partition the rows. Thus a physical update atom is relevant to one field and one base row. Within a lane the invariant coordinates of a backward copy are

$$\chi_{\mathrm{ch}} = (x-t, y, P+vt). \tag{16}$$

For fixed χ_{ch} the end position determines the time and provider location. Its finite source segment is $[L_d, R_d-1]$ in the x coordinate. Successive true ruler intervals on that row are disjoint. Hence each atom belongs to at most one gate's source. Restricting supports preserves disjointness. □

For a selected input $n \geq 1$ put $a = n + n_0$ and

$$I_n = \left(\frac{2C}{\ell_a}, \frac{21C}{10\ell_a} \right), \qquad W = \lfloor C/\varepsilon \rfloor \quad (\varepsilon \in I_n). \tag{17}$$

The parameters guarantee $a \geq k+3 \geq 4$, $\varepsilon \leq C/100$, $0 < C < .01$, and $B \geq 20/C$. In particular

$$\ell_{a-1} < W, \qquad \ell_a \geq 2W. \tag{18}$$

W is used only in the analysis; the automaton does not store it.

Lemma 6.2 (Intrinsic tails)**.** *Define the rational numbers*

$$u_B = 2^{-\lfloor 3BC/100 \rfloor}, \qquad z_B = 2^{-\lfloor BC/3 \rfloor}, \qquad w_B = 2^{-\lfloor BC^2/50 \rfloor}, \qquad \widehat{\eta}_B = \frac{3u_B}{1-4z_B} + 2w_B.$$

For the selected B, $4z_B < 1$. In any W-time bin on one base characteristic, the probability that some finite gate has an intrinsic bad event is at most $\widehat{\eta}_B$. At a single critical gate of depth $a-1$ or a the corresponding probability ρ_B is also at most $\widehat{\eta}_B$.

Proof. For $d = a - 1 - r_{\mathrm{off}}$ the conservative support bound is

$$\begin{aligned} \varepsilon(\ell_d + 1) &< \frac{21}{40} C 4^{-r_{\mathrm{off}}} \frac{2 \cdot 4^a}{2 \cdot 4^a - 1} \\ &\leq \frac{21}{40} \frac{128}{127} C 4^{-r_{\mathrm{off}}} < .53 C 4^{-r_{\mathrm{off}}}. \end{aligned}$$

The displayed rounding estimate uses $a \geq 3$. The exact ℓ_d-atom bound is smaller. For independent Bernoulli variables of mean at most $p_{r_{\mathrm{off}}} = .53C4^{-r_{\mathrm{off}}}$, the exponential moment inequality $1 - p + pe^{\vartheta} \leq \exp(p(e^{\vartheta} - 1))$ and the Chernoff method [5] give

$$\mathbb{P}\{\text{at least } B\theta \text{ successes}\} \leq \exp[-B(\theta \log(\theta / p_{r_{\mathrm{off}}}) - \theta + p_{r_{\mathrm{off}}})], \qquad \theta = .75C.$$

This derivation also applies to the higher integer threshold s. At $r_{\mathrm{off}} = 0$ its exponent exceeds $.03BC$: $\log(75/53) \geq 2(75-53)/(75+53) = 11/32$ and $.75(11/32) - .22 > .03$. The exponent increases by at least $BC/3$ per rank step because $.75 \log 4 - .3975 > 1/3$. At most $3 \cdot 4^{r_{\mathrm{off}}}$ short ends occur in a bin, since their spacing is 4^{d+1} and $W < 4^a$. The sum is at most $3u_B/(1-4z_B)$, using $e^{-x} \leq 2^{-\lfloor x \rfloor}$.

For a long ruler, $\varepsilon W \geq C - \varepsilon \geq .99C$, so (15) gives

$$p_W \geq .98(1 - e^{-.99C}) > .9C.$$

The last inequality follows from $1 - e^{-x} \geq x - x^2/2$ and $C \leq .01$. Also $s/B \leq .75C + 1/B \leq .8C$. Hoeffding's inequality [18, Theorem 2] gives the lower-tail bound $e^{-2B(.1C)^2} = e^{-BC^2/50}$. At most two long ends occur in a bin. Their contribution is at most $2w_B$. This proves both assertions. □

Partition each characteristic (16) into bins $i_{\mathrm{bin}} W \leq t < (i_{\mathrm{bin}} + 1)W$. Write $R(\chi_{\mathrm{ch}}, i_{\mathrm{bin}})$ for the event that some raw flag occurs on that characteristic in the bin. On a phase-zero base characteristic write $S(\chi_{\mathrm{ch}}, i_{\mathrm{bin}})$ for some intrinsic finite-gate failure in that bin, and put $S = \varnothing$ on other rows. This gives one source label $(\chi_{\mathrm{ch}}, i_{\mathrm{bin}})$ for each bin.

Lemma 6.3 (Joint source bound)**.** *For any g distinct labels,*

$$\mathbb{P}\{R \cup S \text{ at all } g \text{ labels}\} \leq \gamma^g, \qquad \gamma = \sqrt{C} + \widehat{\eta}_B^{1/6}. \tag{19}$$

Proof. Raw bins use disjoint atoms and have probability at most $\varepsilon W \le C$. A statistical bin uses its B private lane characteristics and at most its own and preceding bins. Distinct base characteristic labels have disjoint lane characteristics by Lemma 6.1. On each characteristic bins in one residue class modulo three have disjoint source supports. A largest class therefore bounds an intersection of g_S statistical events by $\widehat{\eta}_B^{g_S/3}$. For g_R distinct raw events, Cauchy–Schwarz gives the mixed bound $C^{g_R/2}\widehat{\eta}_B^{g_S/6}$, without assuming that raw and statistical events are independent. Sum over the two source-type choices at each label to obtain (19). □

7 Memory when the computation does not halt

Fix a correct cutoff chart from Proposition 4.3 and one of its two global bit lifts. Start data and the second coordinate at that chart, cloned over providers. Initialize each helper by its ideal chart-driven copying process with independent negative-time noise. Since every point of a lane has an earlier reset or an earlier replacement almost surely, this stationary initialization exists. Spatially average the joint chart/helper law; its phase is Haar and typical almost surely. The positive-time noise is independent of this initial law.

Call a data value or driver bad when it differs from the reference live letter, including when it is bottom. If an ordinary forecast is bad, at least $q - h + 1 = K + 1$ checked providers fail to give the correct letter. Each such check has a bad Moore input. They occupy at least two clusters. Thus a bad forecast admits a two-cluster bad support. The same assertion holds at each virtual iteration of a bad deep driver.

Lemma 7.1 (Error decomposition)**.** *Every finite-time data or second-coordinate error has a finite acyclic derivation whose terminal sources are raw or statistical bins, whose branching steps are ordinary or virtual two-cluster supports, and whose remaining links follow one base characteristic for a lag at most W. Each such characteristic link is followed by a positive support step before another source-free link is needed.*

Proof. A replaced output is a raw source. At an unflagged output, a bad ordinary A_t supplies its two-cluster support. If A_t is correct, the second coordinate is correct. A data error must then be an erasure. A bad current Q_t, including bottom, supplies a depth-T bad support. A correct current Q_t can erase only at a genuine gate of the chart.

At a short gate with correct flag one, its true origin lies less than W updates earlier. If every ordinary and deep driver since that origin is correct, the reset followed by copying and raw marks gives exactly the ideal upper experiment. An erasure therefore requires its intrinsic short failure. Otherwise use a bad A or Q on this base characteristic as the next error. At a long gate with correct flag zero, the true origin is outside the recent W window. If all drivers there are correct, each last HIGH mark forces a singleton one, whatever the older helper state. Absence of the intrinsic long failure gives at least s such singleton fields, preventing an erasure. Again a false erasure supplies a recent driver error.

The B physical lanes share the same driver base at every lag by (4); there is one possible characteristic of driver errors. A bottom ordinary instruction counts as an ordinary error, and a bottom deep instruction as a deep error. Each leads to a two-cluster support. Recursing decreases physical time, or decreases the current finite virtual level before decreasing physical time. Before time zero data drivers may be taken as the perfect chart for this initialization convention. Thus the derivation is finite and acyclic, and a characteristic link always lands on a driver with a support step. □

Lemma 7.2 (Counting the memory witnesses)**.** *The error event at a prescribed site and time is contained in a union of at most K_{mem}^{g} events for each $g \geq 1$, each requiring g distinct source labels. The constants K_{mem} in Section 9 depend on geometry and T, and are independent of B, W, n, n_0 and R.*

Proof. Use the finite derivation before any projection. For an actual $D_s(u, P)$ assign characteristic provider coordinate $P + vs$, planar coordinate $(u_x - s, u_y)$ and bin $\lfloor s/W \rfloor$. For virtual $F_R^j(D_s)(u, P)$ use $P + vs + jv$, planar coordinate $(u_x - s - 1, u_y)$, that bin and its level j. A recentered Q root has exactly its output's characteristic coordinates. Every positive support displacement in provider coordinates is $C_i + o - v$, $o \in \{0, \ldots, 15\}^4$. Its planar displacement is Moore, with an additional $\mathbf{e}_x$ on the final step to actual data. Characteristic links have zero provider and planar characteristic displacement and bin difference at most two.

Use potentials $\psi_0 = 5\sum \chi_{P,i}$, $\psi_i = -5\chi_{P,i}$. Their good and worst gains are 85 and -715. Terminal leaves with one source label have exactly equal potentials, even at unequal physical times. Lemma 5.1 gives at most $48(g-1)$ positive edges. Suppress each maximal source-free characteristic corridor to a corridor link and its entry link. By Lemma 7.1, the next nonterminal reached has a positive support step; charge the two links to that step, once along each copied arm. Thus there are at most two unary entry/corridor links per positive support edge, plus at most $5g$ terminal source-role links. The total is less than $149g < 256g$.

Project the selected witness to characteristic coordinates, bin, virtual level and finite role, and take the union of repeated edges. It remains connected and keeps g distinct marked source labels. Its degree is bounded by the B-independent Δ in Lemma 9.1, because the fixed helper and gate row shifts cancel. Doubled-edge traversals and source subsets give the claimed count there. The charge bound was applied before projection, so projected cycles cause no issue. □

Theorem 7.3 (Two stationary laws)**.** *If $U(n)$ does not halt, then for every $\varepsilon \in I_n$ the perturbation has two distinct spatially translation-invariant stationary laws.*

Proof. Condition on the correct chart. Lemmas 6.3 and 7.2 bound every data or second-coordinate error, uniformly in time, by

$$\sum_{g \geq 1} (K_{\mathrm{mem}}\gamma)^g < \frac{1}{4},$$

where the inequality follows from the parameter choices in Section 9. Integrate over the invariant chart/helper initial law. In the bit-b initialization the probability of the live data cylinder with bit b is greater than $3/4$ at every site and time. The two bit cylinders are disjoint.

The configuration space is compact and the kernel is Feller, since updates are local. A weak limit of temporal Cesaro averages is stationary: the difference of an average and its one-step image is the initial law minus the final law, divided by the averaging length. The initial law and dynamics are spatially invariant, so each limit is invariant under all six spatial coordinates. The one-site cylinder inequalities persist in the limit. The limits from bits zero and one are distinct. □

8 Forgetting when the computation halts

8.1 A missed erasure in a complete history

Fix n with $U(n)$ halting. For this fixed instance the finite family of residues

$$\phi = \tau \bmod 4^{a+1} \quad \text{in both planar coordinates}$$

determines all critical finite origins, ends, their true rides, and their lane phase. For one such ϕ define a flag/mark event $\mathcal{X}^\phi$ at each critical gate: at depth $a-1$ it is the intrinsic short bad event; at depth a it is the long bad event defined by (15). It is absent elsewhere. By Lemmas 6.1 and 6.2, for distinct points S_0,

$$\mathbb{P}\{\mathcal{X}^\phi \text{ at all points of } S_0\} \le \rho_B^{|S_0|}, \qquad \rho_B \le \widehat{\eta}_B. \tag{20}$$

Lemma 8.1 (Missed-erasure implication)**.** *In every complete trajectory with an unmarked chart of phase ϕ, an unmarked live data output at a wrong critical end requires $\mathcal{X}^\phi$ at that output or a wrong live deep query at one prescribed base-characteristic lag $0 \le \ell \le W$.*

Proof. Such an output has live ordinary A_t and no erasure. Its Q_t is live because bottom would erase by default. If this live driver is wrong it is the lag-zero alternative. Otherwise it reads the chart's correct critical end and wrong flag.

At a short wrong flag zero, inspect its true origin. In the absence of a wrong live deep instruction along the ride, the actual interval contains the ideal upper bit at every step: a correct origin resets zero; bottom Q inserts zero by union; bottom A gives UNKNOWN; correct nonorigins copy; a raw interval mark contains its upper bit. This argument removes dependence on helpers older than the origin. With no short bad event there are fewer than s upper ones. Every actual singleton one must be one of these, so $l_w < s$ and the flag-zero erasure fires.

At a long wrong flag one, absence of its bad event gives at least s last HIGH marks in the recent W window. Each writes singleton one. Its true origin is outside the window. Correct nonorigin copying, bottom-Q union and bottom-A UNKNOWN all preserve the possibility of one. In the absence of a wrong live deep instruction these s fields still allow one, giving $h_w \ge s$ and an erasure.

Both contradictions prove the implication. At lag ℓ the nominal query is at $(w - \ell \mathbf{e}_x, P + \ell v, t - \ell)$, with the fixed provider recentering in (3). Thus all lanes give the same query location, and there are at most $W+1$ lag choices. The proof is simultaneous over complete trajectories, with no stationarity or independence of their charts assumed. □

8.2 Packing the historical queries

Lemma 8.2 (Excluding unmarked liveness)**.** *Almost surely every live data or second-coordinate reading in every complete trajectory lies in the same raw set $\mathcal{M}$.*

Proof. Otherwise Lemma 5.3 gives a full chart. Proposition 4.3 gives a wrong critical end w. Its planar site is in the ancient union, so an unmarked live data witness occurs there at some time; a second-coordinate witness supplies a data witness one update earlier. An unmarked live data root has a checked centre parent outside its sole possibly marked cluster. Continue such parents indefinitely at the same planar w.

At a sampled vertex of this path apply Lemma 8.1. If it supplies $\mathcal{X}^\phi$, move one path step backwards. If it supplies a wrong deep query, attach its contour from Lemma 5.4 and restart

at the path vertex one time before its earliest flag. The selected time slabs are disjoint. Let f be their distinct raw flags, m_0 their distinct intrinsic events, and c their contours. Each contour has at least eleven flags, so $c \le f/11$.

Only the next sampled endpoint of a skipped path segment needs recording. Put $V = L + m - 1 = 175$. A gap of h_* ordinary centre steps has net provider displacement in $[0, Vh_*]^4$, with at most $(Vh_* + 1)^4$ possibilities. For a contour with f_i flags, its lag has at most $2W$ choices and its restart gap at most $52 f_i W$. Hence lag, gap and net endpoint together cost at most

$$2[54(V+1)]^5 f_i^5 W^6. \tag{21}$$

The recentering is a fixed translation. A $\mathcal{X}^\phi$ step costs at most q choices. Contour traversals, compositions of f, interleaving and source annotations are bounded in Lemma 9.1; the total count is at most

$$K_{\mathrm{hist}}^{f+m_0} W^{6c} \le K_{\mathrm{hist}}^{f+m_0} W^{6f/11}.$$

The two event families can share raw atoms. Hölder's inequality and (20) give the valid joint weight $\varepsilon^{3f/4} \rho_B^{m_0/4}$. Since $W \le C/\varepsilon$, their counted sum has bases

$$K_{\mathrm{hist}} C^{6/11} \varepsilon^{9/44}, \qquad K_{\mathrm{hist}} \rho_B^{1/4}, \qquad \frac{3}{4} - \frac{6}{11} = \frac{9}{44} > 0. \tag{22}$$

Both bases are below $1/64$ by Section 9. A path segment of length N is covered only if

$$N \le [50 + (W+1)/11] f + m_0.$$

For fixed W, its length tending to infinity therefore forces $f + m_0 \to \infty$. The geometric tail excludes the infinite path. Fix ϕ for the whole argument and then take the finite union of these null events over phases. Countability handles the possible root site, time and provider. Thus the assertion holds simultaneously over all complete histories. □

8.3 Rare births of live proposals

Define the minimal scalar field, separately at every planar u, by increasing limits from zero in the finite past:

$$W_t^{\mathrm{live}}(u, P) = E_t(u, P) \vee \Theta_h(W_{t-1}^{\mathrm{live}})(u, P),$$

where Θ_h means at least h of the q centre providers. This definition has no planar Moore neighbours. Set $N_t^{\mathrm{live}} = \Theta_h(W_t^{\mathrm{live}})$ and $M = h - K = 3K = 196608$.

Lemma 8.3. *Actual liveness in every complete trajectory is contained in W^{live}, and a live ordinary proposal at time t is contained in N_{t-1}^{live}. The latter is dominated by a minimal two-cluster field whose seeds require at least M fresh flags. For g distinct seed groups its joint seed probability is at most*

$$(N_0 \varepsilon^M)^{g/16}, \qquad \lambda = L + m = 176, \quad N_0 = \binom{(2\lambda)^4}{M}. \tag{23}$$

Proof. An unflagged live output has h live data centre parents. By Lemma 8.2 all nodes of its complete live ancestry are in $\mathcal{M}$. An infinite branch would contradict Lemma 5.2. Finite branching then makes the ancestry finite, with flag leaves; induction places its root in W^{live}. The same parent count places a live A_t in N_{t-1}^{live}.

Since $W_t^{\mathrm{live}} = E_t \vee N_{t-1}^{\mathrm{live}}$, a birth with at most $M-1$ fresh provider flags has at least $h - (M-1) = K + 1$ providers in N_{t-1}^{live}, occupying two clusters. Its only alternative seed is M fresh flags. This recursion holds from every finite empty past and hence in the minimal limit.

Partition provider space into cubes $Y = \lambda z + [0, \lambda - 1]^4$. All flag centres for a seed rooted anywhere in Y lie in $\lambda z + [0, 2\lambda - 2]^4$. Declare one group at (u, Y, t) when that box contains at least M flags, and seed the entire root cube. Its probability is at most $N_0 \varepsilon^M$ by a subset union. Equal-parity cubes have disjoint enlarged boxes. Different planar sites or times also have disjoint atoms. A largest one of the sixteen parity classes gives (23). The resulting two-cluster closure is centre-only. Its terminals are these group events. □

Lemma 8.4 (Full-state discrepancies)**.** *For the parameters in Section 9, any two complete full-state trajectories driven by the same noise coincide almost surely.*

Proof. A flagged output is common. An unflagged differing output requires a live $A_t(w - j\mathbf{e}_y)$ in one run for some $0 \le j < B$: otherwise data and second coordinate are bottom and every helper proposal is the same UNKNOWN by (4). Thus every discrepancy lies in a fixed finite union of planar translates of the birth field at time $t - 1$. It also has a discrepant parent in the actual neighbourhood, since equal parents give equal deterministic outputs. A discrepancy would therefore generate an infinite backward path in this shifted union.

Attach grouped contours for that rare field greedily along the path, restarting before the earliest group time. Shift every path and contour vertex down by one time unit to align the birth field with the discrepancy time. Connect a path root to its selected planar target by one literal offset edge. These edges have zero pole increment. Leaves in one group share planar site, time and provider cube, with total potential oscillation at most

$$\Omega = 2 \cdot 5 \cdot 4 \cdot 175 = 7000.$$

By Lemma 5.1 the support and covered path spans each cost at most $H_0 g$, where $H_0 = A_0 + 7000/85 = 11000/85 < 130$. Group identifications cost at most $5g$ star edges and shifted roots at most g connectors. Thus fewer than $282g < 512g$ edges suffice. The selected time slabs make all group identities distinct.

The augmented graph has fixed degree Δ_{rare} as derived in Lemma 9.1; it includes actual dependency offsets, group stars, and the B literal planar shifts. Keeping these as edge labels gives at most K_{rare}^g witnesses. By Lemma 8.3 each has weight at most $(N_0 \varepsilon^M)^{g/16}$. A path length tending to infinity forces $g \to \infty$, by the span bound. The geometric series has base below $1/64$ by Section 9 and excludes every such infinite path, simultaneously at every root. This proves full-state equality. □

Lemma 8.5 (Pullback and attraction)**.** *If complete common-noise trajectories are almost surely unique, then every finite-window pullback image is eventually a singleton almost surely. The PCA has a unique stationary law attracting every initial law.*

Proof. For a fixed good two-sided noise realization, the images

$$K_j = \Psi_{-j,0}(\Sigma^{\mathbb{Z}^6})$$

are nested nonempty compact sets. Their intersection consists precisely of the time-zero states of complete trajectories. To see the nontrivial direction, choose histories from successively earlier starting times and use product compactness to preserve each finite update constraint; then extend forwards. Uniqueness makes the intersection a singleton. Projecting nested compact sets onto any finite window commutes with intersection. The finite projected sets must therefore become singletons after some finite random time.

The failure event is determined by a finite noise cone and decreases to probability zero. Time homogeneity makes its probability the same all-start forward coupling bound. Existence

of a stationary law follows from Feller compactness as in Theorem 7.3. Coupling that stationary start to any initial law bounds their finite-window total variation distance by the failure probability. This proves uniqueness and attraction. □

9 Effective constants and the main theorem

All searches below are over integers or rationals with decidable tests and specify finite constants.

Put

$$\begin{aligned} \Delta &= 10^6(T+2)(9q+10), & K_{\mathrm{mem}} &= 2^{20}(2\Delta^2)^{257}, \\ V &= 175, & K_{\mathrm{hist}} &= 2^{40}[100(V+1)]^{10}(2\Delta^2)^{100}, \\ C &= (128K_{\mathrm{mem}})^{-2}. \end{aligned} \tag{24}$$

Choose the least $k \geq 1$ with $B = 4^k \geq 20/C$, $4z_B < 1$, and

$$\widehat{\eta}_B < \min\{(64K_{\mathrm{mem}})^{-6}, (64K_{\mathrm{hist}})^{-4}\}. \tag{25}$$

This terminates by Lemma 6.2, since its three exponential factors tend to zero. Set $s = \lceil 3CB/4 \rceil$, take κ from Proposition 2.1, and put

$$\begin{aligned} \lambda = 176, \quad M = 196608, \quad N_0 &= \binom{(2\lambda)^4}{M}, \\ \Delta_{\mathrm{rare}} &= 10^6[\kappa + B + (2\lambda)^4 + q + T + 1], \\ K_{\mathrm{rare}} &= 2^{20}(2\Delta_{\mathrm{rare}}^2)^{513}. \end{aligned} \tag{26}$$

Lemma 9.1 (Literal encoding bounds)**.** *The constants in* (24) *and* (26) *dominate the memory, raw-path, historical and rare-group witness counts used above.*

Proof. The projected memory graph records actual data/second-coordinate roles, T virtual levels, three source-hub roles and two entry/corridor roles. With neutral or one of five charge labels there are at most $6(T+7) \leq 32(T+2)$ roles. Support offsets have at most $9q$ Moore/provider choices for ordinary or final virtual edges and another $9q$ for nonfinal virtual edges. Entry, corridor and hub edges have zero spatial characteristic displacement. Allow ten such zero-offset types and five bin differences. Allow every destination role and both orientations. The undirected degree is at most

$$2 \cdot 32(T+2) \cdot 5(18q+10) \leq 640(T+2)(9q+10) < \Delta.$$

A connected embedded edge set with e edges can be encoded by a deterministic traversal of its doubled edges, of length at most $2e$, and a marked source subset. Its count is bounded by $\Delta^{2e}2^{e+1}$. Summing $e \leq 256g$ gives

$$2\sum_{e=0}^{256g}(2\Delta^2)^e \leq 4(2\Delta^2)^{256g} \leq K_{\mathrm{mem}}^g.$$

The slack factor covers root and role conventions. This proves Lemma 7.2; the $101f$ raw-path encoding in Lemma 5.2 fits the same bound.

For a raw contour with f flags, traversal, flag subset and root role cost at most X^{51f} with $X = 2\Delta^2$. Its duration is at most $50(f-1)$. The historical lag and restart endpoint envelope is (21). Since $f^5 \le 2^{5f}$, and $H_* = 100(V+1)$,

$$2[54(V+1)]^5 f^5 \le 2^{6f} H_*^{5f}.$$

Thus one attached contour costs at most $(2^6 H_*^5 X^{51})^f W^6$. Size compositions cost at most 2^f; interleaving with m_0 intrinsic restarts at most 2^{f+m_0}; binary endpoint/source annotations at most $2^{4(f+m_0)}$. Each intrinsic restart has $q \le \Delta \le X$ choices. Their product is bounded by $(2^{10}X)^{f+m_0}$, leaving the displayed $K_{\mathrm{hist}}^{f+m_0} W^{6c}$ as a majorant. The phase is fixed for the whole witness.

For rare groups the support and covered path counts are each at most $H_0 g$; group stars and planar connectors add at most $6g$. Hence $E < (2H_0+6)g < 282g < 512g$. A finite role set for support, path, root, group, port, source and charges fits in 64 roles. Actual path edges have degree at most κ; ordinary supports at most $18q$ including Moore decorations; group-member edges at most $2(2\lambda)^4$; shifted-root edges at most $2B$. A supergraph connecting all roles on all these offsets has degree at most

$$64[2\kappa + 2B + 2(2\lambda)^4 + 18q + 2(T+1)] < \Delta_{\mathrm{rare}}.$$

Use the same doubled-edge traversal and source-subset count with $512g$ in place of $256g$. It is at most K_{rare}^g. Every offset is a literal edge label, including group and planar shifts. □

Define the positive rational threshold

$$\begin{aligned}\varepsilon_* = \min\{&C/100,\ (64K_{\mathrm{mem}})^{-4},\ (64K_{\mathrm{hist}})^{-44},\\ &[64K_{\mathrm{rare}}(N_0+1)]^{-16}\}.\end{aligned} \tag{27}$$

Finally choose the least $n_0 \ge k+2$ such that

$$\frac{21C}{10(2\cdot 4^{n_0+1}-1)} < \varepsilon_*. \tag{28}$$

Only after this choice construct P and R.

Proposition 9.2 (Parameter verification)**.** *The construction is noncircular, the searches terminate, all intervals* (17) *lie in* $(0,\varepsilon_*)$*, and every convergent-series bound used in Sections 7 and 8 holds throughout those intervals.*

Proof. The constants preceding B depend only on the fixed geometry and T. The search for B terminates as shown after (25). The dependency counts depend on B, geometry and T, with no palette-size dependence. Thus ε_* is determined before n_0 or the machine states. Search (28) terminates since its left side tends to zero. The phase decoder terminates by the compactness argument following (9). Each step is a finite effective construction.

The memory bound follows from

$$K_{\mathrm{mem}}\sqrt{C} = 1/128, \qquad K_{\mathrm{mem}}\widehat{\eta}_B^{1/6} < 1/64,$$

so $K_{\mathrm{mem}}\gamma < 1/32$ and its geometric sum is below $1/4$. The raw-path base $K_{\mathrm{mem}}\varepsilon$ is small under (27). The historical statistical base is below $1/64$ by (25). Its other base is at most $K_{\mathrm{hist}}C^{6/11}(64K_{\mathrm{hist}})^{-9} < 1/64$. Finally, since $\varepsilon \le 1$ and $M \ge 1$,

$$K_{\mathrm{rare}}(N_0\varepsilon^M)^{1/16} \le \frac{N_0^{1/16}}{64(N_0+1)} < 1/64.$$

This also ensures that the seed weight is below one.

For $n \geq 1$, (28) places I_n below ε_*. Also $a \geq k+3 \geq 4$, which supplies the $a \geq 3$ premise in Lemma 6.2. The lower endpoint of I_n gives $\ell_a > 2C/\varepsilon \geq 2W$. Its upper endpoint gives $C/\varepsilon > 10\ell_a/21$. As $\ell_a = 4\ell_{a-1} + 3$, this last number exceeds $\ell_{a-1} + 1$, hence $W > \ell_{a-1}$. Thus (18) holds with integer rounding. □

Theorem 9.3. *For the single automaton (6) and the rational intervals (17), for every $n \geq 1$ and every $\varepsilon \in I_n$,*

$$F_\varepsilon \text{ is ergodic} \quad \Longleftrightarrow \quad U(n) \text{ halts.}$$

If $U(n)$ does not halt there are at least two translation-invariant stationary laws. If it halts the unique stationary law attracts every initial law, and every finite-window common-noise pullback image is almost surely eventually a singleton.

Proof. The rule is fixed and computably specified by Propositions 2.1 and 9.2. Nonhalting gives Theorem 7.3. Halting gives Lemmas 8.4 and 8.5. The intervals are nonempty rational open intervals computable from n. Since $\ell_{a+1} = 4\ell_a + 3$, the upper endpoint for $a+1$ lies below the lower endpoint for a; their upper endpoints tend to zero. Thus they are pairwise disjoint and prove Theorem A. □

Corollary 9.4. *For this fixed automaton, deciding ergodicity at a rational noise rate is undecidable, even under the promise that the rate lies in one of the intervals I_n.*

Proof. The midpoint $\varepsilon_n = 41C/(20\ell_{n+n_0})$ is computable and rational. An algorithm deciding the promised question at ε_n would decide whether $U(n)$ halts by Theorem 9.3. □

Corollary B. There is one computably specified translation-invariant finite-range interaction on $\mathbb{Z}^7$, with a finite alphabet and energies in $\{0, 1\}$, and computable pairwise disjoint rational open intervals $\mathcal{J}_n$ of inverse temperature, such that for every $\beta \in \mathcal{J}_n$ there is exactly one Gibbs measure invariant under all translations of $\mathbb{Z}^7$ if $U(n)$ halts, and at least two if it does not. In the halting case this measure is also the only uniformly recurrent, and hence the only periodic, Gibbs measure.

Proof. Use the automaton of Theorem 9.3, write $N_\Sigma = |\Sigma|$, and let N_F be its finite neighbourhood. At $(w, t) \in \mathbb{Z}^6 \times \mathbb{Z}$ put

$$S_{w,t} = \{(w, t)\} \cup \{(w + z, t - 1) : z \in N_F\},$$
$$\Phi_{w,t}(\sigma) = \mathbf{1}\{\sigma(w, t) \neq f((\sigma(w + z, t - 1))_{z \in N_F})\}.$$

These translates define one interaction, independent of β, with finite range and energies $0, 1$. Its Gibbs specification on a finite set Λ uses every factor $e^{-\beta\Phi_{w,t}}$ with $S_{w,t} \cap \Lambda \neq \varnothing$, including factors of exterior children.

Put $\varepsilon(\beta) = N_\Sigma/(e^\beta + N_\Sigma - 1)$. For every parent word η and symbol c,

$$\frac{e^{-\beta\mathbf{1}\{c \neq f(\eta)\}}}{1 + (N_\Sigma - 1)e^{-\beta}} = (1 - \varepsilon(\beta))\mathbf{1}\{c = f(\eta)\} + \frac{\varepsilon(\beta)}{N_\Sigma}.$$

The denominator is independent of the parent word, so normalization cancels it in every finite-volume specification. This is precisely the normalized space–time specification of $F_{\varepsilon(\beta)}$.

The PCA has a finite alphabet, finite range and transition probabilities at least $\varepsilon(\beta)/N_\Sigma > 0$. By the correspondence of [16], in the form of [12, Lemma 2.4 and Corollary 10.4], in every spatial dimension its spatially translation-invariant stationary laws correspond bijectively to

the Gibbs measures of this specification invariant under all space–time translations, through their two-sided stationary trajectory laws. Distinct stationary laws have distinct time-zero marginals and hence distinct trajectory laws. In the halting case the unique stationary law is spatially translation invariant: every spatial translate is stationary and uniqueness identifies it. Its trajectory law is therefore the only Gibbs measure invariant under all translations of $\mathbb{Z}^7$. In the nonhalting case the two translation-invariant laws of Theorem 9.3 give two such Gibbs measures.

Uniform recurrence here is for the full space–time translation action, with syndetic returns to every weak neighbourhood; periodicity means invariance under a finite-index subgroup. On a halting window, Theorem 9.3 gives convergence to the stationary law from every initial law. The strictly positive finite-alphabet finite-range PCA therefore satisfies [12, Theorem 10.10(i)], so its stationary trajectory law is the only uniformly recurrent Gibbs measure, and in particular the only periodic one.

The decreasing computable bijection $\varepsilon \mapsto \log(N_\Sigma/\varepsilon - N_\Sigma + 1)$ sends $I_n = (\varepsilon_n^-, \varepsilon_n^+)$ to

$$\left(\log(N_\Sigma/\varepsilon_n^+ - N_\Sigma + 1), \log(N_\Sigma/\varepsilon_n^- - N_\Sigma + 1)\right).$$

Both endpoints are computable reals, uniformly in n, since the finite alphabet is effectively specified. Approximate them until two rational numbers strictly between them are certified, and let $\mathcal{J}_n$ be the interval between those rationals. The images are disjoint, so the $\mathcal{J}_n$ are disjoint. The correspondence and Theorem 9.3 give the conclusion throughout each $\mathcal{J}_n$. □

Scale and remaining questions. Every window lies below $10^{-5\cdot 10^8}$, and the alphabet has more than $10^{10^{31539}}$ symbols. With the constants above, $k = 52386$, $\log_{10} B = 31539.51470\ldots$ and $n_0 = 888653587$. The rare-birth term in (27) is the minimum. It and the largest window give

$$\log_{10} \varepsilon_* = -535038537.39331\ldots,$$
$$\log_{10} \sup I_1 = -535038537.61745\ldots.$$

Indeed the helper alphabet alone has 100^B symbols, and $\log_{10}(2B) = 31539.81573\ldots > 31539$. The behaviour between windows, a lower-dimensional construction, an exponential mixing bound on halting windows, and the complexity class of the full set of ergodic rates remain questions.

How this work was done

AI note. GPT-6.1 Sol derived the construction and proofs in a long working session in which agents of the same model family audited parts of the argument; Claude Opus 5.5 checked the result refutation-first in three reviews from the other developer. GPT-6.1 Sol also searched the literature and wrote this manuscript. Claude Opus 5.5 then checked the written text refutation-first against the derivation and the three reviews. GPT-6.1 Sol then added Corollary B, which Claude Opus 5.5 checked in a further refutation-first pass. The authors chose the question, directed the work and take responsibility. Models are not listed as authors.

Verification status

The construction and component proofs were derived with separate agents re-deriving and challenging them. The first review, from the other developer, re-derived the contour,

coherence, birth and pullback arguments of Sections 5 and 8 in a general form, including the grouped potential estimate and the complete-history chart argument. The second review checked the fixed-rule assembly, sensor ownership, direct machine, parameter inequalities, feedback memory estimate, historical compression and shifted-target coupling. It identified the missing written proof for finite-board forcing in exceptional phases. The repaired lemma was then re-derived from the rules of [10, Section 4] in the third review. The subsequent refutation-first check of the written text found no mathematical error and identified the credit and presentation corrections made in this version.

The scaffold statements of Proposition 3.1—desubstitution with a unique tower for every tiling, local phase determination, frames, and the finite-rank red-path and ruler-bit lemmas—are formalized in Lean in [10, Appendix D, version 1.2.0], without `sorry`. The formal results include `desub`, `tower_unique`, `phase_local`, `red_col/red_row`, the `rank_bits` lemmas and the frame and laminar lemmas. Their trust base is `propext`, `Classical.choice` and `Quot.sound`. Quarter encoding (`quarter_exists`, `quarter_unique`) is formalized there as well; the typical-phase routing existence is cited from [10, Propositions 5.1 and 5.7 and Lemma 5.3]; the black-type comparison is proved in Section 3. The finite-rank statements used in Lemma 3.3 are among these formalized inputs and apply even when the other axis has infinite rank.

The first review re-derived the pole/frontier argument and grouped estimates. The second and third reviews checked their unequal-time and zero-potential-link use; they did not repeat all of the original fine combinatorics. The check of the written text re-derived the frontier invariant and the numerical estimates. The Wang set R and the physical alphabet were not enumerated; with the constants of Section 9 they are far too large to list. Nonhalting satisfiability follows from the cited typical-phase construction and the finite computations in Proposition 4.3. Corollary B was added after the check of the written text. A further refutation-first check by the other model family re-derived its proof, checked each hypothesis of [12, Lemma 2.4 and Corollary 10.4] for this automaton, and recomputed the scale of Section 9 with an independent program. The additional conclusion about uniformly recurrent and periodic Gibbs measures was checked against [12, Theorem 10.10(i)] in this revision. The new theorem has no formal proof in Lean. The authors' role is selection of the question and direction of the work; no additional human line-by-line verification is claimed.

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